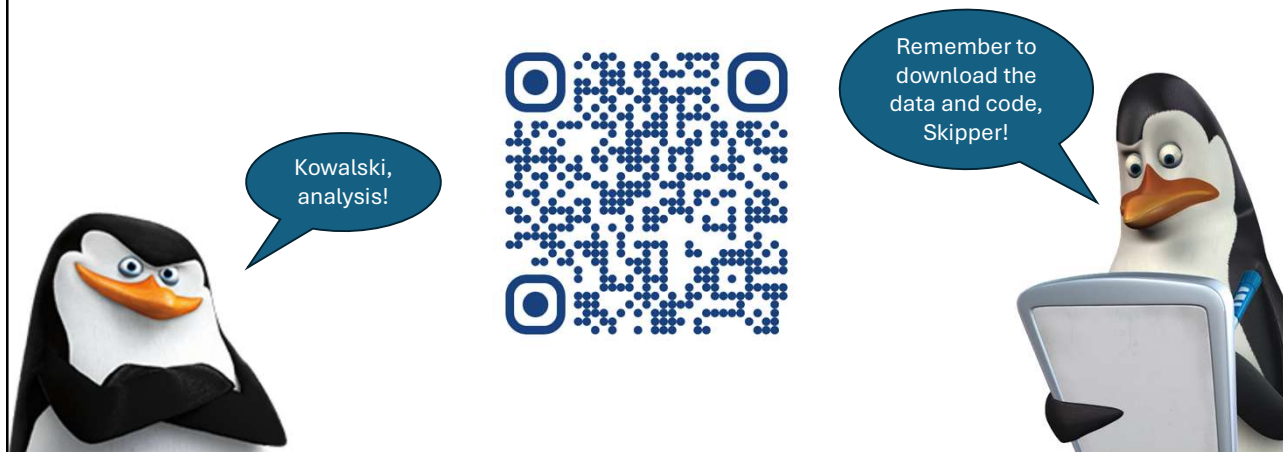


Worked Example

Kaplan-Meier Survival Curves & Cox PH Models



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Before we get started

Data structure for survival analysis

- **single row per subject**
- single **time** variable measuring T time to event (or censoring time, time of last observation)
- a **status** variable coding whether the subject experienced the event or not (censored)
- variables for **covariates** assumed to be time-constant in this structure



Variables

- **time** survival or censoring time
- **Status**
0 = censored,
1 = event **OR**
1 = censored,
2 = event
- **x** covariates

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Data Format

Species	Strain	Insecticide	AI	Metric	Rep	Sex	N	Trauma	T1	T2	T3	T4
Ae. aegypti	IICC G24	res	BA/DTF	Mortality	A	f	34	0	14	8	1	2
Ae. aegypti	IICC G24	res	BA/DTF	Mortality	B	f	25	0	12	7	0	2
Ae. aegypti	IICC G24	res	BA/DTF + ↑ Mort									3
Ae. aegypti	IICC G24	res	BA/DTF + ↑ Mort									1
												-
species	strain	insecticide	ai	rep	t_val	status						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		1	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		2	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		2	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		2	1						
Ae. aegypti	IICC G24	res	BA/DTF + ↑ A		2	1						

time: t_val

status: status

(0 = censored, 1 = event)

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Example: *Helicoverpa zea* survival

Scenario: *Helicoverpa zea* feeding on insecticide-treated plant material. Survival was assessed daily. Insecticide residue determined after experiment was terminated.

Data:

- **population:** Florida, Texas
- **time:** Survival time in days until event or censoring.
- **status:** 0 = censored, 1 = event (death)
- **residue:** Insecticide residue (mg/cm²) on the plant material.



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H. zea survival - Getting started

```
## 1.1 Load libraries ----
# base libraries
# remember to install libraries if you have not used them before!
library(tidyverse)
library(survival) # core survival analysis function

# libraries for data visualizaiton
library(survminer) # recommended for visualizing survival curves
library(ggsurvfit) # alternative for plotting time-to-event endpoints
```



R Code:
Section #1

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H. zea survival - Getting started

```
## 1.2 Load and investigate dataset ----
# Helicoverpa zea feeding on insecticide-treated plant material
dat <- read.csv("Survival Lecture/helicoverpa_survival.csv")
```

```
head(dat)
```

```
> head(dat) # returns the first few rows from the data
```

	population	time	status	residue
1	Texas	14	1	0.9709
2	Texas	1	1	0.9625
3	Texas	5	1	0.9502
4	Florida	1	1	0.9423
5	Texas	5	1	0.8250
6	Florida	1	1	0.8079

```
> str(dat)
```

```
'data.frame': 55 obs. of 4 variables:
```

time: Survival time in
days until event or
censoring

status: 0 = censored, 1
= event (death)

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H. zea survival – Survival Object

Surv() function from 'survival' package

- Creates a **Surv** object with time-to-event data.
- Basic structure: **Surv(time, event)**

```
Surv(time = dat$time, event = dat$status)
```

```
Surv(time, status)
```

Censored observations are marked with +



R Code
Section: #2

```
> Surv(dat$time, dat$status)
[1] 14 1 5 1 5 1 1 2 8 4 1 1 3 9 2 2 5 3
1 7 1 2 1 6 20+ 3 8+ 1 5
[30] 2 2 12+ 15+ 10 4 3 10 3 2 4 4 1 11 8 8 2 14
11 17+ 20+ 6 2 14 12 7
```

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H. zea survival – KM survival curves



R Code
Section: #3

survfit() function from 'survival' package

- generates Kaplan-Meier estimates for survival over time
- Basic structure: **survfit(Surv(time, event) ~ variable)**

```
## 3.1 Complete cohort ----
# Create simple survival curve
s1 <- survfit(Surv(time, status) ~ 1, data = dat)
summary(s1)
```

calculates survival
curve for entire cohort

```
Call: survfit(formula = Surv(time, status) ~ 1, data = dat)
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	55	11	0.8000	0.0539	0.7010	0.913
2	44	9	0.6364	0.0649	0.5211	0.777

One line for each
timepoint for which
events were recorded

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H. zea survival – KM survival curves

```
# short summary of survival curve
s1
```

```
      n events median 0.95LCL 0.95UCL
[1,] 55      49      4        3        7
```

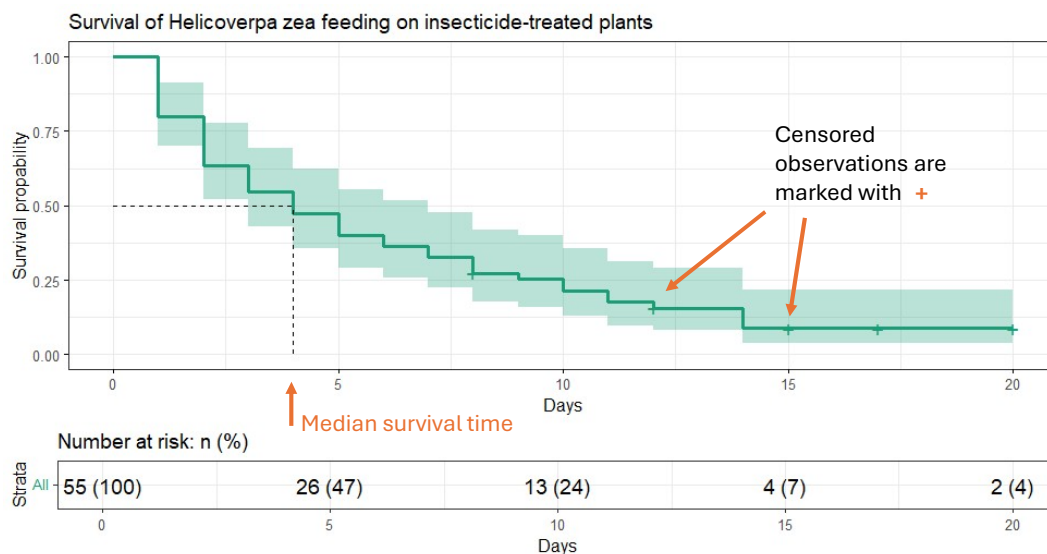
Estimated survival can be used to calculate **probability of survival** beyond **specific times**:

```
# "surv": probability to not have experienced the event at time x
summary(s1, times = c(3, 5, 10, 15, 20))$surv
```

```
[1] 0.5454545 0.4000000 0.2142857 0.0890538 0.0890538
```

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H. zea survival – KM survival curves



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H. zea survival – Population Differences

```
## 3.2 Population ----
s2 <- survfit(Surv(time, status) ~ population, data = dat)
summary(s2)
```

Summary tables are now given for each population

```
Call: survfit(formula = Surv(time, status) ~ population, data = dat)
```

```
population=Florida
time n.risk n.event survival std.err lower 95% CI upper 95% CI
1    27      5    0.8148  0.0748    0.68071    0.975
```

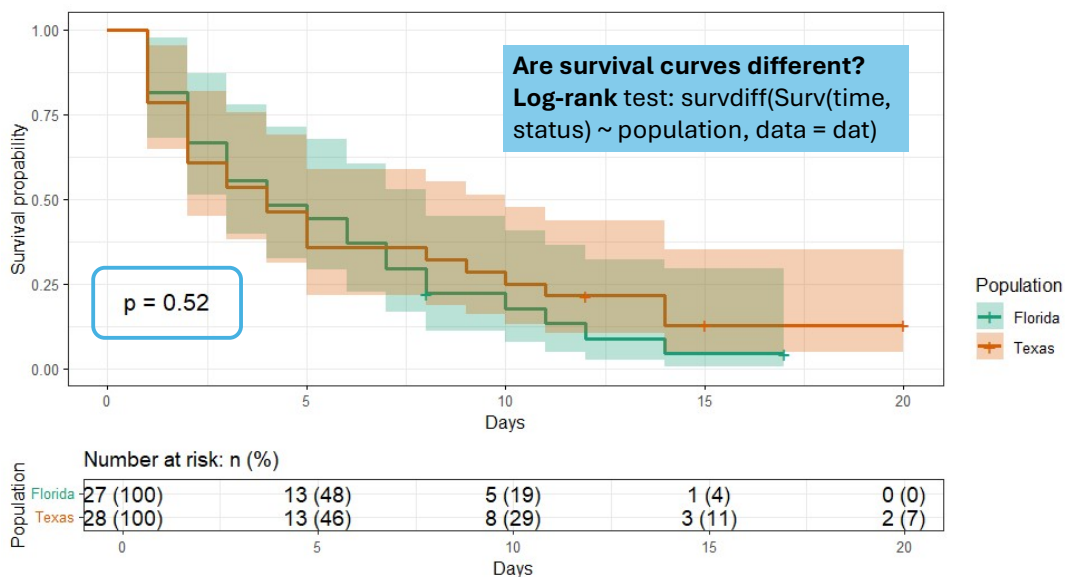
Short summary
of **s2**

	n	events	median	0.95LCL	0.95UCL
population=Florida	27	25	4	3	8
population=Texas	28	24	4	2	10



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H. zea survival – KM survival curves



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R Code
Section: #4

H. zea survival – Cox PH Model

coxph() function from 'survival' package

- Can fit a model with multiple covariates, estimates hazard
- Basic structure: `coxph(Surv(time, event) ~ variable)`

```
cox_m2 <- coxph(Surv(time, status) ~ population, data = dat)
```

```
> cox_m2
```

```
Call:
```

```
coxph(formula = Surv(time, status) ~ population, data = dat)
```

	coef	exp(coef)	se(coef)	z	p
populationTexas	-0.1725	0.8416	0.2887	-0.597	0.55

```
Likelihood ratio test=0.36 on 1 df, p=0.5502
```

```
n= 55, number of events= 49
```

exp(coef): HR of group (TX) relative to reference (FL)

HR = 0.842, or 84.2% of the hazard for FL, i.e. 15.8% reduction in hazard (ns!)

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H. zea survival – Cox PH Model

coxph() function from 'survival' package

- Can fit a model with multiple covariates, estimates hazard
- Basic structure: `coxph(Surv(time, event) ~ variable)`

```
cox_m2 <- coxph(Surv(time, status) ~ population, data = dat)
```

	exp(coef)	exp(-coef)	lower .95	upper .95
populationTexas	0.8416	1.188	0.4779	1.482

Concordance= 0.502 (se = 0.044)

Likelihood ratio test= 0.36 on 1 df, p=0.6

Wald test = 0.36 on 1 df, p=0.6

Score (logrank) test = 0.36 on 1 df, p=0.5

Concordance: measure of predictive accuracy (0.5 = as good as random)

Global model tests assess model fit

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H. zea survival – Changing the Reference

Reference by default:
first group listed for each variable
For categorical variables:
determined by alphabetical order

To change the reference to x:

- 1 Turn variable into a factor
- 2 Redefine levels and assign reference with `relevel(..., ref="x")`



```
dat2 <- dat %>%
  mutate(population = as.factor(population))
# investigate levels in new data set
levels(dat2$population)
# refine the levels using the 'relevel()' function with ref.
# e.g., set "Texas" as reference:
dat2$population <- relevel(dat2$population, ref = "Texas")
```

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H. zea survival – Cox PH Model

Does insecticide residue influence survival?

```
## 4.3 Does insecticide residue influence survival? ----
cox_m3 <- coxph(Surv(time, status) ~ residue, data = dat)
cox_m4 <- coxph(Surv(time, status) ~ population + residue, data = dat)
cox_m5 <- coxph(Surv(time, status) ~ population*residue, data = dat)
```

Based on your **research question** and **study system** you may want to keep covariates in your model **even if n.s.**

Use `anova()` to assess the **contribution of each term**

```
> anova(cox_m5)
Analysis of Deviance Table
Cox model: response is Surv(time, status)
Terms added sequentially (first to last)
```

	loglik	Chisq	Df	Pr(> Chi)
NULL	-160.36			
population	-160.18	0.3570	1	0.550188
residue	-156.18	8.0085	1	0.004656 **
population:residue	-154.49	3.3851	1	0.065790 .

or to **compare** different cox **models**

```
> anova(cox_m4, cox_m5)
Analysis of Deviance Table
Cox model: response is Surv(time, status)
Model 1: ~ population + residue
Model 2: ~ population * residue
```

	loglik	Chisq	Df	Pr(> Chi)
1	-156.18			
2	-154.49	3.3851	1	0.06579 .

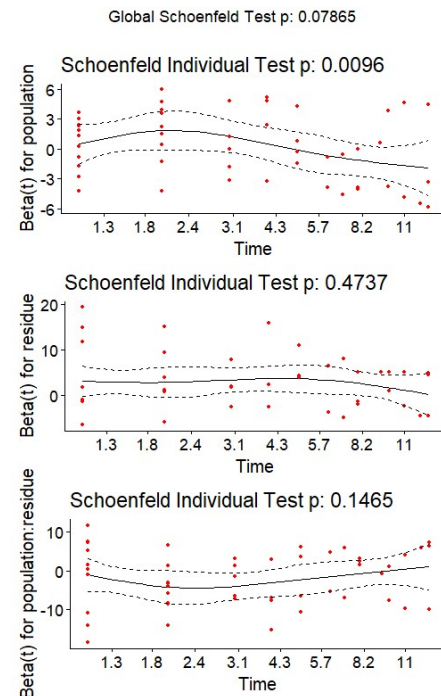
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H. zea survival – Cox PH Assumptions

1. Non-informative censoring ✓
2. Proportional hazard ✗
3. Linearity (continuous covariates)

```
# use the cox.zph() function from the survival package:
test.ph <- cox.zph(cox_m5)
test.ph
# Plot the Schoenfeld residuals over time for each covariate
ggcoxzph(test.ph)
```

```
> test.ph
              chisq df      p
population    6.714  1 0.0096
residue       0.513  1 0.4737
population:residue 2.108  1 0.1465
GLOBAL        6.797  3 0.0786
```



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H. zea survival – Violation of PH assumption

To resolve:

- stratification (categorical variables)
- adding a covariate*time interaction

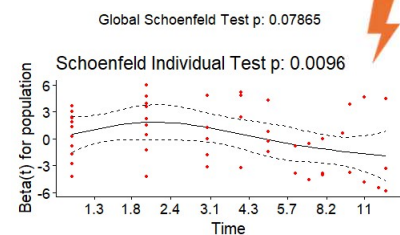
```
cox_m5b <- coxph(Surv(time, status) ~ strata(population)*residue, data = dat)
```

```
cox.zph(cox_m5b)
```

```
              chisq df      p
residue       0.0601  1 0.81
strata(population):residue 0.0426  1 0.84
GLOBAL        0.0612  2 0.97
```

```
      loglik  Chisq Df Pr(>|Chi|)
NULL      -127.07
residue   -122.64 8.8729  1  0.002894 **
strata(population):residue -119.15 6.9762  1  0.008260 **
```

Contribution of stratified variable not assessed



Use stratification only for confounding variables that are not investigated for main effect!

Stratification allows base hazard h_0 to differ between groups

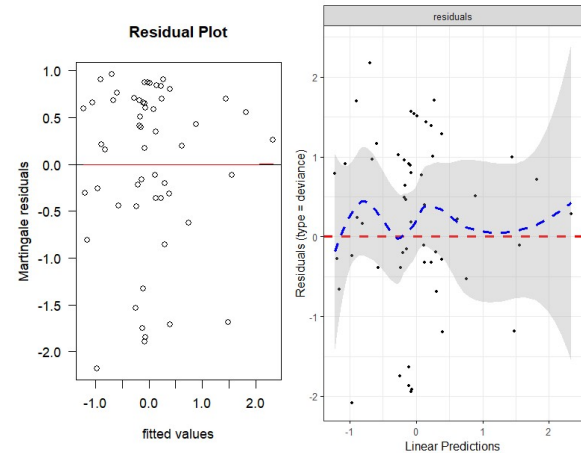
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H. zea survival – Linearity of Covariates

1. Non-informative censoring ✓
2. Proportional hazard ✓
3. Linearity ✓
of continuous covariate

```
# checking for linearity using MARTINGALE residuals
plot(predict(fit), residuals(fit, type="martingale"),
     xlab="fitted values", ylab="Martingale residuals",
     main="Residual Plot", las=1)
abline(h=0)
lines(smooth.spline(predict(fit),
                      residuals(fit, type="martingale")), col="red")

# Alternatively: visual assessment of linearity using DEVIANCE residuals
ggcoxdiagnostics(fit,
                 type="deviance",
                 ox.scale="linear.predictions")
```



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H. zea survival – Interpretation

- 1 **Residue** effect on FL population: HR ≈ 45
→ very strong effect

For each 1-unit increase in residue, the hazard of death increases by a factor of ~ 45



```
> summary(cox_m5b)
Call:
coxph(formula = Surv(time, status) ~ strata(population) * residue,
      data = dat)

n= 55, number of events= 49
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
residue	3.8179	45.5079	1.0069	3.792	0.00015 ***
strata(population)Texas:residue	-3.1056	0.0448	1.1968	-2.595	0.00946 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
residue	45.5079	0.02197	6.32419	327.4684
strata(population)Texas:residue	0.0448	22.32323	0.00429	0.4677

```
Concordance= 0.678 (se = 0.05 )
Likelihood ratio test= 15.85 on 2 df, p=4e-04
Wald test = 15.59 on 2 df, p=4e-04
Score (logrank) test = 18.29 on 2 df, p=1e-04
```

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H. zea survival – Interpretation

- 1 **Residue** effect on FL population: HR $\approx$ 45.5
→ very strong effect
- 2 **Residue** mod. for TX: HR $\approx$ 0.045
→ weaker effect in Texas

Net effect of residue on Texas population:

$$\beta_{res(TX)} = \beta_{res(FL)} + \beta_{TX:res}$$

$$3.8179 + (-3.1056) \approx 0.7123$$

→ HR: $\exp(0.7123) \approx 2.04$



```
> summary(cox_m5b)
Call:
coxph(formula = Surv(time, status) ~ strata(population) * residue,
      data = dat)

n= 55, number of events= 49
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
residue	3.8179	45.5079	1.0069	3.792	0.00015 ***
strata(population)Texas:residue	-3.1056	0.0448	1.1968	-2.595	0.00946 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
residue	45.5079	0.02197	6.32419	327.4684
strata(population)Texas:residue	0.0448	22.32323	0.00429	0.4677

Concordance= 0.678 (se = 0.05)

Likelihood ratio test= 15.85 on 2 df, p=4e-04
Wald test = 15.59 on 2 df, p=4e-04
Score (logrank) test = 18.29 on 2 df, p=1e-04

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H. zea survival – Interpretation

- 1 **Residue** effect on FL population: HR $\approx$ 45
→ very strong effect
- 2 **Residue** mod. for TX: HR $\approx$ 0.045
→ weaker effect in Texas
- 3 **Model performance**
concordance: $\sim$ 0.678
→ moderate predictive accuracy



```
> summary(cox_m5b)
Call:
coxph(formula = Surv(time, status) ~ strata(population) * residue,
      data = dat)

n= 55, number of events= 49
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
residue	3.8179	45.5079	1.0069	3.792	0.00015 ***
strata(population)Texas:residue	-3.1056	0.0448	1.1968	-2.595	0.00946 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
residue	45.5079	0.02197	6.32419	327.4684
strata(population)Texas:residue	0.0448	22.32323	0.00429	0.4677

Concordance= 0.678 (se = 0.05)

Likelihood ratio test= 15.85 on 2 df, p=4e-04
Wald test = 15.59 on 2 df, p=4e-04
Score (logrank) test = 18.29 on 2 df, p=1e-04

*** similar, LRT more robust for smaller sample sizes

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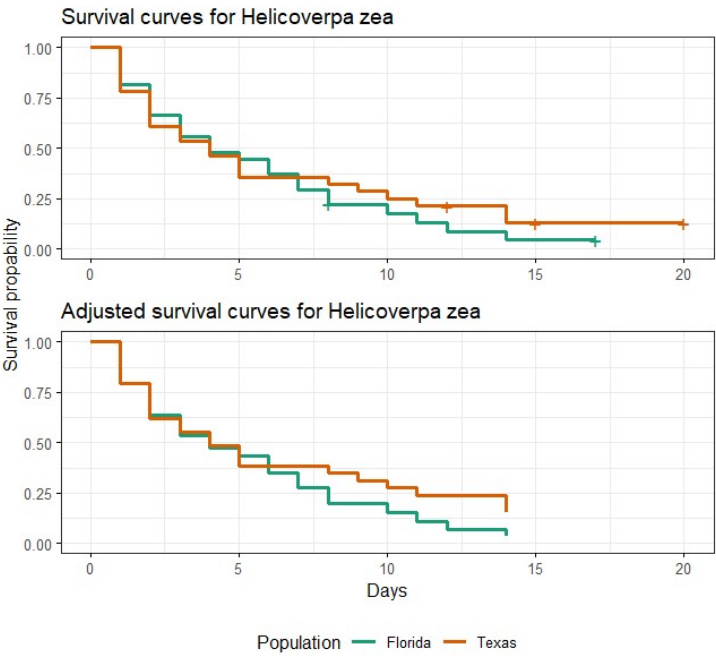
H. zea survival – Adjusted Survival

Adjusted survival curves present expected survival curves for **each sub-populations**



Covariate	HR [†]	95% CI [†]	p-value
residue	45.5	6.32, 327	<0.001
Population(Texas):Residue	0.04	0.00, 0.47	0.009

[†] HR = Hazard Ratio, CI = Confidence Interval



R Code
Section: #5